Haoming Liu

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EDUCATION

National University of Singapore **Present**
M.sc. in Robotics Singapore
Sun Yat-sen University (SYSU) **Sep. 2022 – Jun. 2026**
B.Sc. in Opto-Electronics Information Science and Engineering (GPA: 3.7/5.0) Guangzhou, CN

- **Relevant Coursework:** Digital Electronic Technology (96)//Signals and Systems (91)//Experiments in Electronic Technology (94)//Experiment on Integrated Design of Electronic Circuit (95)//Experiments on Optical Communication Networks (94)// Professional Experiments on Optical Information I (93)

RESEARCH EXPERIENCE

Multimodal Efficient Power Conversion Technology for Electric Vehicles **Oct 2023 – Sep 2025**

- Designed and developed a high-efficiency DC-DC converter based on Buck-Boost and charge-pump topologies, supporting four operating modes: boost charging, buck charging, V2G boost grid feedback, and V2G buck grid feedback.
- Built PSIM simulation models to validate circuit topology and parameter design; completed hardware development including gate-driver design, PCB soldering & debugging, and thermal management — achieving 100V→250V boost and 400V→200V buck at up to 500W.
- Filed 2 invention patents (published); project recognized as a university-level Outstanding Innovation & Entrepreneurship Project

Omnidirectional Intelligent Vehicle Based on Mecanum Wheels **Mar. 2024 – Jul. 2024**

- System architecture: Designed a distributed upper/lower control architecture — Raspberry Pi running ROS2 for LiDAR SLAM mapping and global path planning; STM32F103 handling low-level motor servo and sensor acquisition; real-time UART communication between both layers.
- SLAM navigation: Deployed slam_toolbox with LiDAR driver on Raspberry Pi, fusing IMU and wheel odometry for real-time indoor grid-map construction and autonomous localization. Implemented global A*/Dijkstra path planning and local dynamic obstacle avoidance (DWB) via ROS2 Navigation2.
- Low-level execution: Independently completed the lower-computer PCB layout in JLC EDA and PCBA assembly; implemented multi-channel encoder PID closed-loop control in Keil5, with IMU angular-velocity correction for mecanum-wheel heading drift.

Robotic Arm Control Based on SO-ARM101 **Nov 2025 – May 2026**

- ROS 2 Motion Planning & Low-Level Control: Built a low-level arm control system based on ROS 2 and MoveIt 2, using the RRT* algorithm for collision-free Cartesian-space path planning and inverse kinematics.
- VLA Model Deployment & Fine-Tuning: Built a wrist+top dual-view synchronous capture framework and cleaned 50+ expert demonstration trajectories; trained an end-to-end imitation learning policy under the ACT framework, mitigating few-shot overfitting via Chunk Length tuning and multi-view data augmentation; constructed a complete SmolVLA training-and-inference pipeline, achieving a 60% success rate on real-robot pick-and-place tasks.
- Tactile Modality Fusion: Performed hardware calibration and signal preprocessing for a FlexiTac-V2 flexible pressure sensor , building a tactile-visual multimodal data collection pipeline under the LeFlexiTac framework; encoded tactile pressure maps and fused them with visual observations for fine-tuning of SmolVLA, improving real-robot grasping success rate by 30% over a vision-only baseline.

High-DoF Teleoperation System for Contact-Rich Dexterous Manipulation **Jul 2026 – Present**

- Architected a bimanual teleoperation system with two decoupled paths: MANUS gloves to 20-DoF Wuji dexterous hands, and VIVE trackers to Franka via cuRobo IK, for imitation-learning data collection.
- Built the hand pipeline retargeting 21-keypoint glove input to 20-DoF joint angles via analytical-gradient IK, achieving smooth real-time control at ~50 ms latency over ROS 2.

INTERNSHIP

Ronovo Surgical **Jan 2026 – May 2026**
Software Engineering Intern Shanghai, China

- Built a VR teleoperation system for surgical robots using the Pico headset, integrating the full pipeline of XR spatial tracking - DDS real-time communication - stereo-vision video feedback; optimized video-stream hard encoding/decoding and communication protocols to reduce end-to-end latency to 50–100 ms, enabling scalable high-quality expert demonstration data collection.
- MuJoCo dynamics simulation & low-level control optimization: Developed a Jacobian pseudo-inverse + Remote Centre of Motion constrained joint-optimization IK solver with null-space projection for singularity avoidance and redundancy management; incorporated gravity-compensation to eliminate start-up jitter and ensure smooth servo control.

SKILLS & LANGUAGES

- **Computer:** Python, MATLAB, PSIM, Pytorch, IsaacLab, Mujoco
- **Languages:** Mandarin (native), English (fluent, IELTS 7.5), Cantonese (native)